**0580_m23_qp_42_Q2**

**2** **(a)** 100 students take part in a reaction test.
The table shows the results.

| Reaction time (seconds) | 6 | 7 | 8 | 9 | 10 | 11 |
|---|---|---|---|---|---|---|
| Number of students | 3 | 32 | 19 | 29 | 11 | 6 |

**(i)** Write down the mode.

..... s [1]

**(ii)** Find the median.

..... s [1]

**(iii)** Calculate the mean.

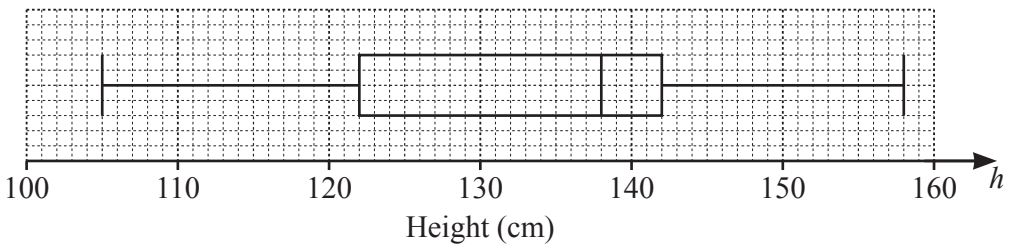
..... s [3]

**(iv)** Two students are chosen at random.

Find the probability that both their reaction times are greater than or equal to 9 seconds.

..... [2]

**(b)** The box-and-whisker plot shows the heights, $h$ cm, of some students.




**(i)** Find the range.

..... cm [1]

**(ii)** Find the interquartile range.

..... cm [1]

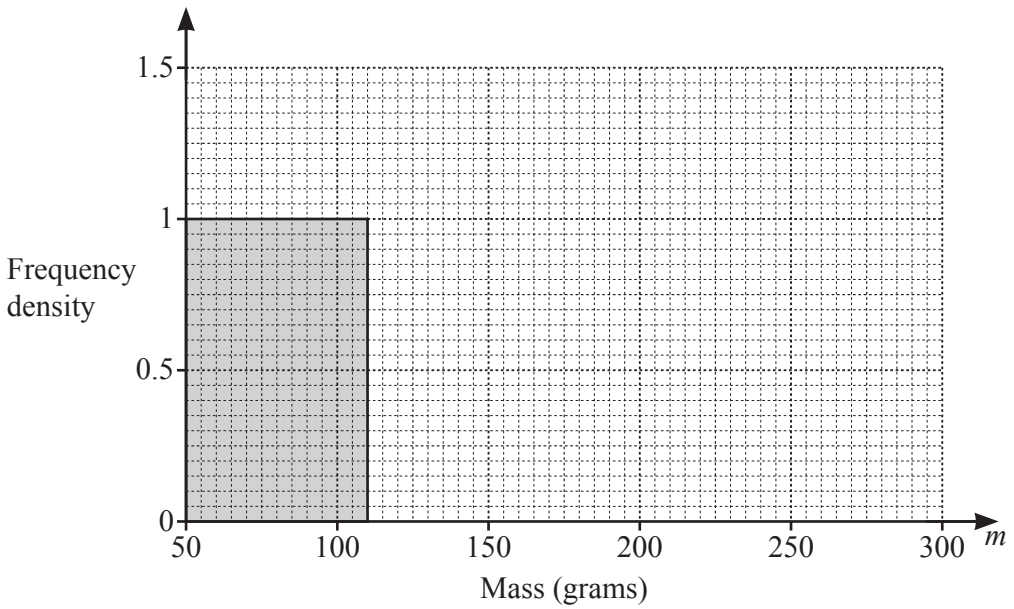
**(c)** The mass of each of 200 potatoes is measured.
The table shows the results.

| Mass ($m$ grams) | $50 < m \leqslant 110$ | $110 < m \leqslant 200$ | $200 < m \leqslant 300$ |
|---|---|---|---|
| Frequency | 60 | 99 | 41 |

**(i)** Calculate an estimate of the mean.

..... g [4]

**(ii)** Complete the histogram to show the information in the table.




[2]